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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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List of Abbreviations

MAS Multi-agent System

STL Signal Temporal Logic

1. Introduction

This report is the deliverable for Task 2.3 in the third year of the SymAware project. It is based on the following research contributions:

- E. E. Vlahakis, L. Lindemann, P. Sopasakis and D. V. Dimarogonas, "Probabilistic Tube-based Control Synthesis of Stochastic Multi-Agent Systems under Signal Temporal Logic," 2024 *IEEE 63rd Conference on Decision and Control (CDC), Milan, Italy*, 2024, pp. 1586-1592, doi: 10.1109/CDC56724.2024.10886279.
- E. E. Vlahakis, L. Lindemann, P. Sopasakis and D. V. Dimarogonas, "Conformal Prediction for Distribution-Free Optimal Control of Linear Stochastic Systems," in *IEEE Control Systems Letters*, vol. 8, pp. 2835-2840, 2024, doi: 10.1109/LCSYS.2024.3514472
- A. B. Kordabad, E. E. Vlahakis, L. Lindemann, S. Gros, D. V. Dimarogonas, and S. Soudjani (2025). "Data-Driven Distributionally Robust Control for Interacting Agents under Logical Constraints." *Submitted to IEEE Transactions on Automatic Control* [Online] https://arxiv.org/abs/2503.09816
- E. E. Vlahakis, L. Lindemann, and D. V. Dimarogonas, "Conformal Data-driven Control of Stochastic Multi-Agent Systems under Collaborative Signal Temporal Logic Specifications," *to be submitted to IEEE 64th Conference on Decision and Control (CDC)*

These works focus on efficient risk quantification methods for control synthesis of complex multi-agent systems (MAS) under stochastic uncertainty and Signal Temporal Logic (STL) specifications and contribute to the objectives of the SymAware project. They advance both theoretical foundations and practical methods for task planning and control in complex, stochastic multi-agent settings where uncertainty information is limited. The purpose and content of this report are outlined in the following section.

1.1 Purpose and contents of this report

This report is the third deliverable of Work Package WP2 on spatiotemporal awareness and specifically on Task 2.3 about task planning for a multi-agent system based on risk awareness. The report is organized as follows: Chapter 2 provides the problem statement that we focus on along with preliminaries and mathematical background for the statistical inference tools we use and the syntax of STL. Chapter 3 presents the proposed uncertainty quantification method that facilitates the multi-agent control synthesis under STL specifications presented in Chapter 4.

Simulations and numerical validations of our approach are also demonstrated in Chapter 4. Conclusions and extensions of the current work are drawn in Chapter 5. Next, we provide a summary of the state-of-the-art and motivation for our work.

1.2 Literature review and motivation

Multi-agent systems (MAS) can be found in many applications, such as robotics, autonomous vehicles, and cyber-physical systems. When these systems are stochastic, the formal specification of system properties can be formulated probabilistically enabling a trade-off between quantifying uncertainty and adjusting feasibility. As the complexity in control synthesis from temporal logics under uncertainty grows with the dimensionality of the overall system, existing approaches typically focus on single-agent [1] or non-stochastic [2,3] systems.

Here, we focus on signal temporal logic (STL) [4] to formally formulate and verify specifications for a wide range of MAS. STL employs predicates coupled with Boolean and temporal operators, allowing precise specification of complex spatio-temporal properties in a dynamical system. In a deterministic setting, it is possible to design sound and complete algorithms that guarantee STL satisfaction [5], based on the quantitative semantics [6]. Here, we consider stochastic MAS and formulate the underlying task planning problem as a chance-constrained optimal control problem, where the goal is to satisfy a multi-agent STL specification with a desired probability.

Probabilistic guarantees play a crucial role in many MAS applications that involve stochastic disturbances and safety. As chance-constrained problems are generally non-convex and computationally intractable, most approaches solve deterministic relaxations by applying constraint-tightening techniques based on available information about uncertainty [7]. Constraint tightening for probabilistic satisfaction has been studied when the underlying probability distribution is known [8], in cases of bounded disturbances [9], or using probabilistic reachable sets [10–12]. These approaches may be restricted to Gaussian settings, involve computationally costly operations, or often rely on the multivariate Chebyshev inequality, which can lead to conservative bounds. In contrast, sampling-based methods such as [13–15] can be significantly more flexible in relaxing chance-constrained problems by utilizing available disturbance samples, termed scenarios, and can provide formal guarantees based on scenario optimization (SO) [16,17].

To address stochasticity in the STL framework, the works in [18–20] propose risk constraints over predicates while preserving Boolean and temporal operators. Probabilistic STL in [21] allows one to express uncertainty by incorporating random variables into predicates, while [22] introduces chance-constrained temporal logic for modeling uncertainty. Similar approaches are found in [23,24]. Top-down approaches imposing chance constraints on the entire specification are explored in [1,25,26]. Although important, these works focus on low-dimensional systems and lack guidance on extending to MAS [27]. A recent extension of [1] to stochastic MAS under STL [28] considers only a single joint task per agent and bounded distributions.

An alternative distribution-free framework for streamlined uncertainty quantification, equipped with formal guarantees, is provided by conformal prediction (CP). Originally introduced by Vovk and Shafer [29,30], CP uses a calibration dataset to infer prediction regions for a test dataset with a specified probability, while remaining distribution-agnostic. CP has been applied in machine learning [31], chance-constrained optimization [32], and control for safety verification and planning [33–35], see [36] for a recent survey.

Leveraging its distribution independence, we apply CP to tackle an optimal task planning problem for stochastic MAS under STL specifications, assuming data-driven noise information. Our uncertainty quantification approach is closely related to [14, 15], which employ offline sampling for constraint tightening and probabilistic reachable set computation. Both methods assume a fixed feedback control policy, which preserves convexity in the underlying scenario-based relaxed programs, as required by SO in [16, 17] to enable solutions with a priori guarantees. Disturbance feedback parameterization [37] is employed in [13] for online SO-based stochastic optimal control, which maintains convexity and is well-suited for short control horizons if the optimization is implemented online. In our approach, we address optimal task planning for stochastic MAS under collaborative STL specifications with data-driven noise information. Using the notion of cliques from graph theory, we define collaborative tasks for arbitrary agent groups, enabling a compositional structure of the multi-agent STL formula. Given disturbance sample datasets for each agent, we use CP to provide a distribution-free solution to the underlying chance-constrained STL synthesis problem. We introduce nonconformity scores as functions of prediction regions (PRs) of aggregate error trajectories, capturing uncertainty in collaborative tasks. We develop an iterative PR-scaling and disturbance-feedback synthesis approach bounding training error trajectory samples, which are subsequently calibrated on separate datasets to ensure probabilistic guarantees via CP. The stochastic optimal control problem is then relaxed by tightening the robustness functions of collaborative tasks based on their Lipschitz constants and the computed error bounds. For multi-agent STL specifications, this is a less conservative alternative to tightening approaches relying on union-bounding or the STL structure [1, 12, 28].

Finally, exploiting the compositional structure of the multi-agent STL formula, we propose a distributed model-predictive-control-like algorithm, where agent-level problems are solved in a distributed fashion according to a graph-informed scheduling policy based on the interaction graph induced by the underlying multi-agent STL structure, improving scalability.

2. Preliminaries & Problem statement

2.1 Notation

The sets of real numbers and nonnegative integers are $\mathbb{R}$ and $\mathbb{N}$, respectively. Let $N \in \mathbb{N}$. Then, $\mathbb{N}_{[0,N]} = \{0, 1, \ldots, N\}$. Let $x_1, \ldots, x_n$ be vectors. Then, $x = (x_1, \ldots, x_n) = [x_1^\intercal \ \cdots \ x_n^\intercal]^\intercal$. We denote by $\boldsymbol{x}(a : b) = (x(a), \ldots, x(b))$ an aggregate vector consisting of $x(t)$, $t \in \mathbb{N}_{[a,b]}$, representing a trajectory. When $x(t)$, $t \in \mathbb{N}_{[a,b]}$, are random vectors, $\boldsymbol{x}(a : b) = (x(a), \ldots, x(b))$ is a random process. Let $x_i(t)$, for $t \in \mathbb{N}_{[0,N]}$ and $i \in \mathbb{N}_{[1,M]}$. Then, $\boldsymbol{x}(0 : N) = (x(0), \ldots, x(N))$ denotes an aggregate trajectory when $x(t) = (x_1(t), \ldots, x_M(t))$, $t \in \mathbb{N}_{[0,N]}$. A random variable (vector) w following a distribution $\mathcal{D}_w$ is denoted as $w \sim \mathcal{D}_w$ and the expected value of w is $\mathbb{E}(w)$. The probability of event Y is $\Pr\{Y\}$. Quantile$(\{\mathscr{D}\}, \delta)$ is the δ-th quantile of a distribution $\mathscr{D}$, i.e., for $Z \sim \mathscr{D}$, Quantile$(\{\mathscr{D}\}, \delta) = \inf\{z : \Pr\{Z \leq z\} \geq \delta\}$. The ceiling operator is $\lceil \cdot \rceil$. The cardinality of a set $\mathcal{V}$ is $|\mathcal{V}|$. The symbol $\otimes$ denotes the Kronecker product.

2.2 Statistical tools and risk metrics

2.2.1 Conformal Prediction

Let $\mathcal{R}^{(0)}, \ldots, \mathcal{R}^{(k)}$ be samples of an independent and identically distributed (i.i.d.) random variable $\mathcal{R}$. We will refer to $\mathcal{R}^{(\varsigma)}$, $\varsigma \in \mathbb{N}_{[0,k]}$, as a nonconformity score. Given a failure probability $\theta \in (0, 1)$, one wishes to obtain a bound $q \in \mathbb{R}$ for $\mathcal{R}^{(0)}$, which we call test point, so that

$$\Pr\left\{\mathcal{R}^{(0)} \leq q\right\} \geq 1 - \theta, \tag{2.1}$$

where q is computed from the samples $\mathcal{R}^{(1)}, \ldots, \mathcal{R}^{(k)}$, which we call calibration dataset. Specifically, q may be attained as $q = \text{Quantile}(\{\mathcal{R}^{(1)}, \ldots, \mathcal{R}^{(k)}, \infty\}, 1 - \theta)$, which is the $(1 - \theta)$th quantile of the empirical distribution $\{\mathcal{R}^{(1)}, \ldots, \mathcal{R}^{(k)}, \infty\}$. Assuming $\mathcal{R}^{(1)} \leq \cdots \leq \mathcal{R}^{(k)}$, one can pick $q = \mathcal{R}^{(p)}$, where $p = \lceil (k + 1)(1 - \theta) \rceil$, which indicates the pth smallest nonconformity score. Note that q is finite with $p \in \mathbb{N}_{[1,k]}$ if $k \geq \lceil (k + 1)(1 - \theta) \rceil$. This choice of q ensures that (2.1) holds since test point $\mathcal{R}^{(0)}$ and calibration data $\mathcal{R}^{(1)}, \ldots, \mathcal{R}^{(k)}$ are i.i.d. [38]. This is summarized below.

Lemma 1 *[38, Lemma 1] If $\mathcal{R}^{(0)}, \ldots, \mathcal{R}^{(k)}$ are i.i.d. random variables, then for any $\theta \in (0, 1)$, we have*

$$\Pr\left\{\mathcal{R}^{(0)} \leq \text{Quantile}(\{\mathcal{R}^{(1)}, \ldots, \mathcal{R}^{(k)}, \infty\}, 1-\theta)\right\} \geq 1 - \theta. \tag{2.2}$$

Remark 1 *The coverage guarantees in (2.2) are marginal as the probability is defined over the randomness in the draw of test and calibration points* $\mathcal{R}^{(0)}, \mathcal{R}^{(1)}, \ldots, \mathcal{R}^{(k)}$*. Conditional coverage guarantees of the form* $\Pr\{\mathcal{R}^{(0)} \leq C \mid \mathcal{R}^{(1)}, \ldots, \mathcal{R}^{(k)}\}$ *are unfortunately not possible to obtain. However, one can show that the conditional probability is a random variable following a beta distribution centered at* $1-\theta$ *regardless of* k *[31, 36]. Notably, probably approximately correct coverage guarantees* $\Pr_c\{\Pr\{\mathcal{R}^{(0)} \leq \text{Quantile}(\{\mathcal{R}^{(1)}, \ldots, \mathcal{R}^{(k)}, \infty\}, 1-\hat{\theta})\} \geq 1-\theta\} \geq 1-\beta$ *can be obtained by setting* $\hat{\theta} = \theta - \sqrt{\frac{\ln(1/\beta)}{2k}}$*, with the "outer" probability* $\Pr_c$ *taken with respect to the randomness in the draw of the calibration data* $\mathcal{R}^{(1)}, \ldots, \mathcal{R}^{(k)}$*, and* $\beta \in (0,1)$ *[39].*

2.2.2 Conditional value at risk (CVaR)

For a random variable $\mathcal{R} \sim \mathscr{D}$ and confidence level $(1-\theta)$, $\text{VaR}_{1-\theta}$ is defined as

$$\text{VaR}_{1-\theta}(\mathcal{R}) := \inf\{\eta \in \mathbb{R} \mid \Pr\{\mathcal{R} \leq \eta\} \geq 1-\theta\},$$

that is $\text{VaR}_{1-\theta}(\mathcal{R}) = \text{Quantile}(\{\mathscr{D}\}, 1-\theta)$. Then, one can show that

$$\text{VaR}_{1-\theta}(\mathcal{R}) \leq q \Leftrightarrow \Pr\{\mathcal{R} \leq q\} \geq 1-\theta, \tag{2.3}$$

where a bound q can be obtained in a data-driven fashion as in (2.2). Let $\mathcal{R}(M) : \mathcal{M} \to \mathbb{R}$ be a random variable. Unfortunately, optimizing $\text{VaR}_{1-\theta}(\mathcal{R}(M))$ is challenging since VaR is typically nonconvex in M even if $\mathcal{R}(M)$ is a convex function. Alternatively, CVaR of $\mathcal{R}$ with a confidence level of $(1-\theta)$, denoted as $\text{CVaR}_{1-\theta}(\mathcal{R})$, measures the expected value of $\mathcal{R}$ in the θ-tail exceeding the threshold $\text{VaR}_{1-\theta}(\mathcal{R})$, i.e.,

$$\text{CVaR}_{1-\theta}(\mathcal{R}) := \mathbb{E}\left(\mathcal{R} \mid \text{VaR}_{1-\theta}(\mathcal{R}) \leq \mathcal{R}\right).$$

CVaR can be formulated as

$$\text{CVaR}_{1-\theta}(\mathcal{R}) = \min_{\eta \in \mathbb{R}} \mathbb{E}\left(\eta + \frac{1}{\theta}(\mathcal{R}-\eta)_+\right), \tag{2.4}$$

where $(\cdot)_+ = \max\{0, \cdot\}$ [40]. Since CVaR is a coherent risk measure that satisfies conditions such as convexity and monotonicity [40], it is possible to optimize CVaR instead of VaR, using standard convex and linear programming techniques, relying on the fact that CVaR provides a tight upper bound for $\text{VaR}_{1-\theta}(\mathcal{R})$ [40], that is,

$$\text{VaR}_{1-\theta}(\mathcal{R}) \leq \text{CVaR}_{1-\theta}(\mathcal{R}).$$

Figure 2.1 compares the different risk measures including VaR, CVaR and $\mathbb{E}$.

2.3 Signal temporal logic (STL)

We consider STL formulas with standard syntax

$$\varphi := \top \mid \pi \mid \neg\phi \mid \phi_1 \wedge \phi_2 \mid \phi_1 U_{[t_1,t_2]} \phi_2, \tag{2.5}$$

where $\pi := (\mu(x) \geq 0)$ is a predicate, $\mu(x) : \mathbb{R}^{n_x} \to \mathbb{R}$ is a predicate function of $x \in \mathbb{R}^{n_x}$, and ϕ, ϕ_1, and ϕ_2 are STL formulas, which are built recursively using predicates π, logical operators $\neg$ and

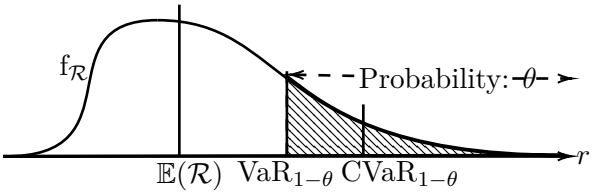

Figure 2.1: Illustration of the different risk measures $\mathbb{E}(\mathcal{R})$, $\mathrm{VaR}_{1-\theta}$ and $\mathrm{CVaR}_{1-\theta}$ for random variable $\mathcal{R}$ with probability density function $f_{\mathcal{R}}$.

$\wedge$, and the *until* temporal operator U, with $[t_1, t_2] \equiv \mathbb{N}_{[t_1,t_2]}$. We omit $\vee$ (*or*), $\Diamond$ (*eventually*) and $\Box$ (*always*) operators from (2.5) and the sequel, as these may be defined by (2.5), e.g., $\phi_1 \vee \phi_2 = \neg(\neg\phi_1 \wedge \neg\phi_2)$, $\Diamond_{[t_1,t_2]}\phi = \top U_{[t_1,t_2]}\phi$, and $\Box_{[t_1,t_2]}\phi = \neg\Diamond_{[t_1,t_2]}\neg\phi$.

Let π be a predicate and ϕ an STL formula. We write $\pi \in \phi$ to indicate that π is part of the formula ϕ. We denote by $\boldsymbol{x}(t) \models \phi$, $t \in \mathbb{N}$, the satisfaction of ϕ, verified over $\boldsymbol{x}(t) = (x(t), x(t+1), \ldots)$. The validity of a formula ϕ can be verified using Boolean semantics: $\boldsymbol{x}(t) \models \pi \Leftrightarrow \mu(x(t)) \geq 0$, $\boldsymbol{x}(t) \models \neg\phi \Leftrightarrow \neg(\boldsymbol{x}(t) \models \phi)$, $\boldsymbol{x}(t) \models \phi_1 \wedge \phi_2 \Leftrightarrow \boldsymbol{x}(t) \models \phi_1 \wedge \boldsymbol{x}(t) \models \phi_2$, $\boldsymbol{x}(t) \models \phi_1 U_{[a,b]} \phi_2 \Leftrightarrow \exists \tau \in t \oplus \mathbb{N}_{[a,b]}$, s.t. $\boldsymbol{x}(\tau) \models \phi_2 \wedge \forall \tau' \in \mathbb{N}_{[t,\tau]}, \boldsymbol{x}(\tau') \models \phi_1$. Based on the Boolean semantics, the horizon of a formula is recursively defined as [4]: $N^{\pi} = 0$, $N^{\neg\phi} = N^{\phi}$, $N^{\phi_1 \wedge \phi_2} = \max(N^{\phi_1}, N^{\phi_2})$, $N^{\phi_1 \, U_{[a,b]} \phi_2} = b + \max(N^{\phi_1}, N^{\phi_2})$.

STL is endowed with quantitative semantics [6]: A scalar-valued function $\rho^{\phi} : \mathbb{R}^n \times \cdots \times \mathbb{R}^n \to \mathbb{R}$ of a signal indicates how robustly a signal $\boldsymbol{x}(t)$ satisfies a formula ϕ. The robustness function is defined recursively as follows:
$\rho^{\pi}(\boldsymbol{x}(t)) = \mu(x(t))$,
$\rho^{\neg\phi}(\boldsymbol{x}(t)) = -\rho^{\phi}(\boldsymbol{x}(t))$,
$\rho^{\phi_1 \wedge \phi_2}(\boldsymbol{x}(t)) = \min(\rho^{\phi_1}(\boldsymbol{x}(t)), \rho^{\phi_2}(\boldsymbol{x}(t)))$,
$\rho^{\phi_1 U_{[a,b]} \phi_2}(\boldsymbol{x}(t)) = \max_{\tau \in t \oplus \mathbb{N}_{[a,b]}} (\min(Y_1(\tau), Y_2(\tau')))$,
with $Y_1(\tau) = \rho^{\phi_1}(\boldsymbol{x}(\tau))$, $Y_2(\tau') = \min_{\tau' \in \mathbb{N}_{[t,\tau]}} \rho^{\phi_2}(\boldsymbol{x}(\tau'))$, π being a predicate, and ϕ, ϕ_1, and ϕ_2 being STL formulas.

To facilitate the definition of joint STL formulas for agents involved in collaborative tasks, we borrow the concept of cliques from graph theory to represent groups of agents engaged in the same STL task. The definition of cliques is provided next.

Definition 1 *Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph potentially containing self-loops and multiple edges with node set $\mathcal{V}$, cardinality $M = |\mathcal{V}|$, and edge set $\mathcal{E}$. Let also $\mathcal{V}' \subseteq \mathcal{V}$, with $|\mathcal{V}'| \geq 1$, and define $\mathcal{E}_{\mathcal{V}'} \subseteq \mathcal{E}$ as the set of edges connecting the nodes $\mathcal{V}'$. Then, $\mathcal{G}' = (\mathcal{V}', \mathcal{E}_{\mathcal{V}'})$ is a clique [41], if $\mathcal{G}'$ is a complete graph. The set of cliques of $\mathcal{G}$ is defined as $\mathcal{K} = \{\nu \subseteq \mathcal{V} \mid (\nu, \mathcal{E}_{\nu})$ is a complete subgraph of $\mathcal{G}\}$.*

Consider a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with clique set $\mathcal{K}$, a clique $\nu \in \mathcal{K}$, with $\nu = (i_1, \ldots, i_{|\nu|})$, and vectors $x_{i_j}(t)$, $j \in \mathbb{N}_{[1,|\nu|]}$, with $t \in \mathbb{N}$, Then, $x_{\nu}(t) = (x_{i_1}(t), \ldots, x_{i_{|\nu|}}(t))$ is an aggregate vector. We denote by $\boldsymbol{x}_{\nu}(t) \models \phi$ the validity of an STL formula defined over the aggregate trajectory $\boldsymbol{x}_{\nu}(t) = (x_{\nu}(t), x_{\nu}(t+1), \ldots)$, which verifies that $\rho^{\phi}(\boldsymbol{x}_{\nu}(t)) \geq 0$.

2.4 Multi-agent setup and problem statement

2.4.1 Multi-agent system

Dynamics

We consider M agents with discrete dynamics

$$x_i(t+1) = A_i x_i(t) + B_i u_i(t) + w_i(t), \tag{2.6}$$

where $x_i(t) \in \mathcal{X}_i \subseteq \mathbb{R}^{n_i}$, $u_i \in \mathcal{U}_i \subseteq \mathbb{R}^{m_i}$, and $w_i(t) \in \mathcal{W}_i \subseteq \mathbb{R}^{n_i}$, $\boldsymbol{w}_i(0:N-1) \sim \mathscr{D}_{w_i}$ are the state, input and disturbance vectors, respectively, the initial condition, $x_i(0)$, is known, (A_i, B_i) is a stabilizable pair, with $A_i \in \mathbb{R}^{n_i \times n_i}$, $B_i \in \mathbb{R}^{n_i \times m_i}$, $i \in \mathbb{N}_{[1,M]}$, and $t \in \mathbb{N}$. Let $\nu = (i_1, \ldots, i_{|\nu|})$, where $i_j \in \mathbb{N}_{[1,M]}$, with $j \in \mathbb{N}_{[1,|\nu|]}$. By collecting individual state, input, and disturbance vectors, as $x_\nu(t) = (x_{i_1}(t), \ldots, x_{i_{|\nu|}}(t)) \in \mathcal{X}\nu \subseteq \mathbb{R}^{n_\nu}$, $u_\nu(t) = (u_{i_1}(t), \ldots, u_{i_{|\nu|}}(t)) \in \mathcal{U}_\nu \subseteq \mathbb{R}^{m_\nu}$, and $w_\nu(t) = (w_{i_1}(t), \ldots, w_{i_{|\nu|}}(t)) \in \mathcal{W}_\nu \subseteq \mathbb{R}^{n_\nu}$, respectively, we write the aggregate dynamics of $|\nu|$ agents as

$$x_\nu(t+1) = A_\nu x_\nu(t) + B_\nu u_\nu(t) + w_\nu(t), \tag{2.7}$$

where $A_\nu = \mathrm{diag}(A_{i_1}, \ldots, A_{i_{|\nu|}})$, $B_\nu = \mathrm{diag}(B_{i_1}, \ldots, B_{i_{|\nu|}})$, and the state, input, and disturbance sets are $\mathcal{X}_\nu = \mathcal{X}_{i_1} \times \cdots \times \mathcal{X}_{i_{|\nu|}}$, $\mathcal{U} = \mathcal{U}_{i_1} \times \cdots \times \mathcal{U}_{i_{|\nu|}}$, and $\mathcal{W} = \mathcal{W}_{i_1} \times \cdots \times \mathcal{W}_{i_{|\nu|}}$, respectively. When $\nu = (1, \ldots, M)$, the aggregate dynamics of the entire MAS are written as

$$x(t+1) = Ax(t) + Bu(t) + w(t), \tag{2.8}$$

where $A = \mathrm{diag}(A_1, \ldots, A_M)$ and $B = \mathrm{diag}(B_1, \ldots, B_M)$. Next, we define ν as a clique indicating a group of agents (or an individual agent) involved in a collaborative (or an individual) STL task.

STL specification

Let $\mathcal{V} = \{1, \ldots, M\}$ denote the set of indices of all agents in MAS. The MAS is subject to a conjunctive STL formula, given by

$$\phi = \bigwedge_{\nu \in \mathcal{K}_\phi} \phi_\nu, \tag{2.9}$$

where each conjunct ϕ_ν follows the syntax in (2.5) and represents a formula that involves a group of agents ν, which we call a clique indicating all agents in ν can interact with each other. The set $\mathcal{K}_\phi$ collects all these cliques induced by ϕ, and may include individual agents ($|\nu| = 1$) or group of agents ($1 < |\nu| \leq |\mathcal{V}|$).

The structure of ϕ in (2.9) induces an interaction graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of nodes, and $\mathcal{E} = \{(\nu_i, \nu_j) \mid \nu_i, \nu_j \in \nu,\ \nu \in \mathcal{K}_\phi\}$ is the set of edges. Note that $\mathcal{E}$ may include self loops (indicating individual tasks) and multiple edges (indicating that two agents may be jointly involved in more than one collaborative task). In what follows, we denote by $\mathcal{K}_\phi$ the set of cliques induced by ϕ, $\mathcal{K}_\phi^i = \{\nu \in \mathcal{K}_\phi \mid \nu \ni i\}$ the subset of cliques containing agent i, and define $\bar{\mathcal{K}}_\phi$, such that $\mathcal{K}_\phi = \bar{\mathcal{K}}_\phi \cup \mathcal{V}$.

Next, let $\pi := (\mu(y) \geq 0)$ be a predicate in ϕ, where $\mu(y) : \mathbb{R}^{n_y} \to \mathbb{R}$. The vector $y \in \mathbb{R}^{n_y}$ can represent either an individual state vector $x_i \in \mathbb{R}^{n_i}$, where $i \in \{1, \ldots, M\}$, or an aggregate state vector $x_\nu \in \mathbb{R}^{n_\nu}$ that collects the states of agents in clique $\nu \in \mathcal{K}_\phi$.

Disturbance

Let $\boldsymbol{w}_i(0{:}N{-}1) = \{w_i(0), w_i(1), \ldots, w_i(N{-}1)\}$ denote the disturbance sequence for the ith agent. We assume that the joint distribution of the random vectors $w_i(t){\in}\mathbb{R}^{n_i}$, $t{\in}\mathbb{N}_{[0,N-1]}$, is unknown, and instead that a disturbance dataset, $\mathcal{D}^{w_i} = \{\boldsymbol{w}_i^{(0)}, \ldots, \boldsymbol{w}_i^{(k)}\}$, of $k+1$ samples is available, with $\boldsymbol{w}_i^{(\varsigma)} = (w_i^{(\varsigma)}(0), \ldots, w_i^{(\varsigma)}(N{-}1))$, $\varsigma \in \mathbb{N}_{[0,k]}$. We assume that the disturbance sequence samples in $\mathcal{D}^{w_i}$ are independent, where each sample represents one realization of the process. We also assume that trajectory samples $\boldsymbol{w}_i^{(\varsigma)}$ are independent agent-wise, that is $\boldsymbol{w}_i^{(\varsigma)}$, $\boldsymbol{w}_j^{(\varsigma)}$ are independent for all $i, j \in \mathbb{N}_{[1,M]}$ and $\varsigma \in \mathbb{N}_{[0,k]}$. Note that although $\boldsymbol{w}_i^{(\varsigma)}$, $\varsigma \in \mathbb{N}_{[0,k]}$, are assumed to be independent, the random vectors $w_i(t)$, $t \in \mathbb{N}_{[0,N-1]}$, may be correlated across time for the ith agent. We partition $\mathcal{D}^{w_i}$ into training and calibration datasets, $\mathcal{D}^{w_i}_{\text{train}} = \{\boldsymbol{w}_i^{(k_1+1)}, \ldots, \boldsymbol{w}_i^{(k)}\}$ and $\mathcal{D}^{w_i}_{\text{cal}} = \{\boldsymbol{w}_i^{(1)}, \ldots, \boldsymbol{w}_i^{(k_1)}\}$, where $k_1 + 1 < k$.

2.4.2 Problem statement

We wish to solve the stochastic optimal control problem

$$\begin{aligned} \underset{\substack{\boldsymbol{u}(0:N-1)\\ \boldsymbol{x}(0:N)}}{\text{Min.}} \; & \mathbb{E}\left(\sum_{i=1}^{M}\Big(\sum_{t=0}^{N-1}(\ell_i(x_i(t), u_i(t))) + V_{f,i}(x_i(N))\right) \\ \text{s.t. } & x(t+1) = Ax(t) + Bu(t) + w(t),\ t \in \mathbb{N}_{[0,N)}, \\ & \Pr\{\boldsymbol{x}(0:N) \models \phi\} \geq 1-\theta, \text{ with } x(0) = x_0, \end{aligned} \tag{2.10}$$

where $\ell_i : \mathbb{R}^{n_i} \times \mathbb{R}^{m_i} \to \mathbb{R}$, $V_{f,i} : \mathbb{R}^{n_i} \to \mathbb{R}$ are costs to be optimized over the optimization variables are $\boldsymbol{u}(0{:}N{-}1) = (u(0), \ldots, u(N-1))$, $\boldsymbol{x}(0{:}N) = (x(0), \ldots, x(N))$, with $u(t) = (u_1(t), \ldots, u_M(t))$, $t \in \mathbb{N}_{[0,N-1]}$, and $x(t) = (x_1(t), \ldots, x_M(t))$, $t \in \mathbb{N}_{[0,N]}$, respectively, ϕ is a multi-agent STL formula, with structure as in (2.9) and syntax as in (2.5), to be satisfied by $\boldsymbol{x}(0{:}N)$ with a probability $1-\theta$, where $\theta \in (0,1)$, x_0 is a known initial condition of the MAS, and N is the horizon of ϕ. Solving the problem directly is challenging due to the probabilistic constraint, the expectation operator in the cost function, the lack of information on the distribution of the disturbance $w(t)$, and the growing complexity with the number of agents. Note that based on the multi-agent STL formula in (2.9), the STL chance constraint in (2.10) is equivalently written as

$$\Pr\{\boldsymbol{x}_\nu(0:N) \models \phi_\nu,\ \forall \nu \in \mathcal{K}_\phi\} \geq 1-\theta. \tag{2.11}$$

3. Data-driven feedback design & uncertainty quantification

3.1 Decomposition of dynamics

Due to the linearity in (2.6), the state of each agent can be decomposed into a deterministic part, $z_i(t)$, and an error, $e_i(t)$, i.e., $x_i(t) = z_i(t) + e_i(t)$. Consider the disturbance-feedback control policy [37]

$$u_i(t) = \sum_{k=0}^{t-1} \Gamma_i^{t,k} w_i(k) + v_i(t), \tag{3.1}$$

which is causal, since the disturbance sequence $\{w_i(0), \ldots, w_i(t-1)\}$ is measurable at time t, with $\Gamma_i^{t,k} \in \mathbb{R}^{\Gamma_i \times n_i}$. Then,

$$z_i(t+1) = A_i z_i(t) + B_i v_i(t), \tag{3.2a}$$

$$e_i(t+1) = A_i e_i(t) + B_i \sum_{k=0}^{t-1} \Gamma_i^{t,k} w_i(k) + w_i(t), \tag{3.2b}$$

where $z_i(0) = x_i(0)$ and $e_i(0) = 0$. Consider a clique $\nu \in \mathcal{K}_\phi$, where $\nu = (i_1, \ldots, i_{|\nu|})$, let

$$\Gamma_\nu^{t,k} = \mathrm{diag}(\Gamma_{i_1}^{t,k}, \ldots, \Gamma_{i_{|\nu|}}^{t,k}), \tag{3.3}$$

and define the aggregate vectors $z_\nu(t) = (z_{i_1}(t), \ldots, z_{i_{|\nu|}}(t))$, $e_\nu(t) = (e_{i_1}(t), \ldots, e_{i_{|\nu|}}(t))$, and $v_\nu(t) = (v_{i_1}(t), \ldots, v_{i_{|\nu|}}(t))$. Then, the aggregate dynamics of the agents in ν can be decomposed into

$$z_\nu(t+1) = A_\nu z_\nu(t) + B_\nu v_\nu(t), \tag{3.4a}$$

$$e_\nu(t+1) = A_\nu e_\nu(t) + \sum_{k=0}^{t-1} \Gamma_\nu^{t,k} w_\nu(k) + w_\nu(t). \tag{3.4b}$$

Given particular disturbance feedback gains $\Gamma_\nu^{t,k}$, $k \in \mathbb{N}_{[0,t-1]}$, $t \in \mathbb{N}_{[1,H-1]}$, the error system in (3.4b) can be analyzed independently of the one in (3.4a). The above control choice will allow us to analyze prediction regions of trajectories or the error systems in (3.4b). We are interested in prediction regions as defined next.

Definition 2 *Let $\boldsymbol{y}(a:b) \in \mathbb{R}^{n(b-a)}$ be a random process, with $y(t) \in \mathbb{R}^n$, $t \in \mathbb{N}_{[a,b]}$. We call the ball $\mathbb{B}(q) \subseteq \mathbb{R}^{n(b-a)}$ a prediction region (PR) of $\boldsymbol{y}(a:b)$ at probability level $1-\theta$, if* $\Pr\{\|\boldsymbol{y}(a:b)\| \leq q\} \geq 1-\theta$.

We aim to determine parameters to tighten the robustness functions of the formulas ϕ_ν, $\nu \in \mathcal{K}_\phi$ guided by the size of PRs obtained for the error systems in (3.4b), in light of the probabilistic constraint in (2.10) or equivalently in (2.11). The following proposition underpins this approach.

Proposition 1 *Let $\boldsymbol{x}(0:N) = \boldsymbol{z}(0:N) + \boldsymbol{e}(0:N)$, with $\boldsymbol{x}(0:N) = (x(0), \dots, x(N))$, $\boldsymbol{z}(0:N) = (z(0), \dots, z(N))$ and $\boldsymbol{e}(0:N) = (e(0), \dots, e(N))$. Let $q > 0$ be such that $\Pr\{\boldsymbol{e}(0:N) \in \mathbb{B}(q)\} \geq 1-\theta$. If $\boldsymbol{z}(0:N) + \boldsymbol{e}(0:N) \models \phi$ for all $\boldsymbol{e}(0:N) \in \mathbb{B}(q)$, then $\Pr\{\boldsymbol{x}(0) \models \phi\} \geq 1-\theta$.*

Proof Define events $Y_x := \boldsymbol{x}(0) \models \phi$, $Y_e := \boldsymbol{e}(0:N) \in \mathbb{B}(q)$, and let Y_e' be the complement of Y_e. From the law of total probability, it follows that $\Pr\{Y_x\} = \Pr\{Y_x|Y_e\}\Pr\{Y_e\} + \Pr\{Y_x|Y_e'\}\Pr\{Y_e'\} \geq 1-\theta$, since by assumption, $\Pr\{Y_x|Y_e\} = 1$ and $\Pr\{Y_e\} \geq 1-\theta$, and $\Pr\{Y_x|Y_e'\}\Pr\{Y_e'\} \geq 0$.

3.2 Error trajectory samples

To facilitate the proposed data-driven design approach, we construct error trajectory samples for each agent, using the available disturbance datasets and the error dynamics in (3.2b) with initial condition $e_i(0) = 0$, $i \in \mathbb{N}_{[1,M]}$. Specifically, let matrices

$$\boldsymbol{A}_i = \begin{bmatrix} I_{n_i} & 0 & \cdots & 0 \\ A_i & I_{n_i} & \ddots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A_i^{N-1} & A_i^{N-2} & \cdots & I_{n_i} \end{bmatrix}, \tag{3.5}$$

$$\boldsymbol{\Gamma}_i = \begin{bmatrix} 0 & \cdots & \cdots & \cdots & 0 \\ \Gamma_i^{1,0} & 0 & \cdots & \cdots & 0 \\ \Gamma_i^{2,0} & \Gamma_i^{2,1} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \Gamma_i^{N-1,0} & \Gamma_i^{N-1,1} & \cdots & \Gamma_i^{N-1,N-2} & 0 \end{bmatrix}, \tag{3.6}$$

and $\boldsymbol{B}_i = \boldsymbol{A}_i(I_N \otimes B_i)$. Then, for the ith agent, a dataset with error trajectory samples $\boldsymbol{e}_i^{(\varsigma)}(1{:}N) = (e_i^{(\varsigma)}(1), \dots, e_i^{(\varsigma)}(N))$ can be constructed from disturbance samples $\boldsymbol{w}_i^{(\varsigma)}(0{:}N-1) \in \mathcal{D}^{w_i}$ as

$$\mathcal{D}^{e_i} = \{\boldsymbol{e}_i^{(0)}(1{:}N), \dots, \boldsymbol{e}_i^{(k)}(1{:}N)\}, \tag{3.7a}$$

$$\boldsymbol{e}_i^{(\varsigma)}(1{:}N) = (\boldsymbol{A}_i + \boldsymbol{B}_i\boldsymbol{\Gamma}_i)\boldsymbol{w}_i^{(\varsigma)}(0{:}N-1),\ \varsigma \in \mathbb{N}_{[0,k]}. \tag{3.7b}$$

The linear dependence of the error trajectories on the gains $\boldsymbol{\Gamma}_i$, $i \in \mathbb{N}_{[1,M]}$, facilitates the disturbance feedback design in parallel with quantification of appropriate PRs for the error trajectories. In the following, we partition $\mathcal{D}^{e_i}$ into $\mathcal{D}^{e_i}_{\text{train}}$ and $\mathcal{D}^{e_i}_{\text{cal}}$, where trajectory samples in $\mathcal{D}^{e_i}_{\text{train}}$ are constructed by disturbance samples in $\mathcal{D}^{w_i}_{\text{train}}$ and are parameterized by the disturbance feedback gains $\boldsymbol{\Gamma}_i$, while trajectory samples in $\mathcal{D}^{e_i}_{\text{cal}}$ are constructed by disturbance samples in $\mathcal{D}^{w_i}_{\text{cal}}$ for fixed disturbance feedback gains. We use the $\mathcal{D}^{e_i}_{\text{train}}$ dataset for feedback design in Sec. 3.3, and the $\mathcal{D}^{e_i}_{\text{cal}}$ dataset to obtain PRs with probabilistic guarantees in Sec. 3.4.

3.3 Data-driven PR scaling and disturbance feedback design

Here, we present a data-driven approach to designing disturbance feedback gains $\mathbf{\Gamma}_i$, $i \in \mathbb{N}_{[1,M]}$, and scaling parameters C_ν, $\nu \in \mathcal{K}_\phi$ to weigh the size of PRs of the trajectories of the error systems in (3.4b) corresponding to agents involved in cliques $\nu \in \mathcal{K}_\phi$, using the training dataset $\mathcal{D}^{e_i}_{\text{train}}$. To this end, we introduce nonconformity scores $E^{(\varsigma)}(C, \mathbf{\Gamma})$, $\varsigma \in \mathbb{N}_{[k_1+1,k]}$, which are parametrized over the disturbance feedback gains in $\mathbf{\Gamma} = \{\mathbf{\Gamma}_1, \dots, \mathbf{\Gamma}_M\}$ and the weights in $C = \{C_\nu\}_{\nu \in \mathcal{K}_\phi}$, defined as

$$E^{(\varsigma)}(C, \mathbf{\Gamma})) = \max_{\nu \in \mathcal{K}_\phi} \left(C_\nu \| \boldsymbol{e}^{(\varsigma)}_\nu(1:N) \| \right), \tag{3.8}$$

where $\boldsymbol{e}^{(\varsigma)}_\nu(1:N) = (e_\nu(1), \dots, e_\nu(N))$, with $\nu = (i_1, \dots, i_{|\nu|})$ and $e^{(\varsigma)}_{i_j}(t)$ defined as in (3.7b), for $j \in \mathbb{N}_{[1,|\nu|]}$. The synthesis of disturbance feedback gains in $\mathbf{\Gamma}$ and weights in C is formulated by the following optimization problem

$$\underset{C,\, \mathbf{\Gamma}}{\text{Minimize}} \ \text{Quantile}(\{E^{(k_1+1)}(C, \mathbf{\Gamma}), \dots, E^{(k)}(C, \mathbf{\Gamma})\}, \hat{\theta}) \tag{3.9a}$$

$$\text{subject to } \mathbf{\Gamma}_i \boldsymbol{w}^{(\varsigma)}_i(0{:}N{-}1) \in \mathcal{U}^N_i, \ \varsigma \in \mathbb{N}_{(k_1,k]}, \tag{3.9b}$$

$$0 \le C_\nu \le 1, \ \sum_{\nu \in \mathcal{K}_\phi} C_\nu = 1, \tag{3.9c}$$

with $\mathcal{U}^N_i = \mathcal{U}_i \times \cdots \times \mathcal{U}_i \in \mathbb{R}^{Nm_i}$, $\mathbf{\Gamma}_i$ as in (3.6), and $\hat{\theta} = (1 + \frac{1}{k-k_1-1})(1-\theta)$, where the confidence $1-\theta$ is replaced by $(1 + \frac{1}{k-k_1-1})(1-\theta)$ (see [36, Sec. 2.1] for details). Despite the linearity of error trajectories in (3.7b) in $\mathbf{\Gamma}_i$, the nonconformity score in (3.8) is not jointly convex in the feedback gains in $\mathbf{\Gamma}$ and the weights in C. Moreover, even when the weights in C are fixed, the $\text{VaR}_{\hat{\theta}}$ cost is not necessarily convex, and the problem in (3.9) can be addressed via an efficient nonlinear solver. A similar data-driven feedback synthesis problem is addressed in our previous work [42, Sec. IV], where a genetic algorithm was successfully employed in a single-agent setting. While the problem in (3.9) could be tackled similarly, we propose a CVaR-based iterative relaxation, which may be beneficial in our multi-agent setting, where complexity increases with the number of agents. Specifically, leveraging the bilinear dependence of the nonconformity scores in (3.8) on the variables in C and in $\mathbf{\Gamma}$, we propose an iterative procedure to optimize sequentially over C and $\mathbf{\Gamma}$ by solving convex problems employing the properties of CVaR. Specifically, we define the problem P as

$$\underset{\substack{\eta \ge 0,\, Y \ge 0 \\ C,\, \mathbf{\Gamma}}}{\text{Minimize}} \ \eta + \frac{1}{q} \sum_{\varsigma=k_1+1}^{k} (Y - \eta)_+ \tag{3.10a}$$

$$\text{subj. to } Y \ge C_\nu \| \boldsymbol{e}^{(\varsigma)}_\nu(1{:}N) \|, \ \forall \nu \in \mathcal{K}_\phi, \ \varsigma \in \mathbb{N}_{(k_1,k]}, \tag{3.10b}$$

$$\mathbf{\Gamma}_i \boldsymbol{w}^{(\varsigma)}_i(0:N-1) \in \mathcal{U}^N_i, \ \varsigma \in \mathbb{N}_{(k_1,k]}, \tag{3.10c}$$

$$0 \le C_\nu \le 1, \ \sum_{\nu \in \mathcal{K}_\phi} C_\nu = 1, \tag{3.10d}$$

where $q = (k-k_1-1)(1-\hat{\theta})$, and the variable Y is introduced to address the max operator in (3.8) by the constraints in (3.10b). Next, we denote by $P(C)$ the optimization in (3.10) for fixed weights in C and by $P(\mathbf{\Gamma})$ the optimization in (3.10) for fixed feedback gains in $\mathbf{\Gamma}$. Algorithm

Algorithm 1 Iterative procedure for solving (3.10)

1: Set C^0=$\{C_\nu$=0 for $|\nu|$>1, C_i=$\frac{1}{M}$ for $i\in\mathbb{N}_{[1,M]}\}$
2: **for** τ in $1:\tau_{\max}$ **do**
3: Solve $P(C^{\tau-1})$ and return $\mathbf{\Gamma}^\tau$
4: Solve $P(\mathbf{\Gamma}^\tau)$ and return C^τ.
5: **return** $(C^* \leftarrow C^{\tau_{\max}},\ \mathbf{\Gamma}^* \leftarrow \mathbf{\Gamma}^{\tau_{\max}})$

1 summarizes a procedure for solving (3.10) iteratively. We note that the optimization problems $P(C)$ and $P(\mathbf{\Gamma})$ solved at each iteration of Algorithm 1 are convex, ensuring that the algorithm can be executed efficiently. While it does not guarantee convergence to a local optimum, since $P(C)$ and $P(\mathbf{\Gamma})$ optimize over one subset of variables at a time, this iterative technique, known as a (block-)coordinate descent algorithm, is a widely used heuristic that performs well in practice.

3.4 Error prediction regions

Given the disturbance feedback gains $\mathbf{\Gamma}^* = \{\mathbf{\Gamma}_1^*, \dots, \mathbf{\Gamma}_M^*\}$ and weights $C^* = \{C_\nu^*\}_{\nu\in\mathcal{K}_\phi}$, obtained either by directly solving the problem in (3.9) or via Algorithm 1, we now derive PRs for the error trajectories of agents in cliques $\nu \in \mathcal{K}_\phi$ with the desired confidence level as follows.

Proposition 2 *Let $\mathbf{\Gamma}^*$ and C^* collect disturbance feedback gains and weights, respectively, which are feasible for the problem in (3.9). Construct the calibration trajectory dataset $\mathcal{D}_{\text{cal}}^{e_i}$ as in (3.7), using the calibration disturbance set $\mathcal{D}_{\text{cal}}^{w_i}$ and the disturbance feedback gains in $\mathbf{\Gamma}^*$, define nonconformity scores*

$$E^{(\varsigma)}(C^*, \mathbf{\Gamma}^*) = \max_{\nu\in\mathcal{K}_\phi} \left(C_\nu^* \|\boldsymbol{e}_\nu^{(\varsigma)}(1:N)\| \right), \tag{3.11}$$

where $\varsigma \in \mathbb{N}_{[0,k_1]}$, and compute

$$q = \text{Quantile}(\{E^{(1)}(C^*, \mathbf{\Gamma}^*), \dots, E^{(k_1)}(C^*, \mathbf{\Gamma}^*), \infty\}, 1-\theta). \tag{3.12}$$

Then,

$$\Pr\left\{ \boldsymbol{e}_\nu^{(0)}(1:N) \in \mathbb{B}(q/C_\nu^*),\ \forall \nu \in \mathcal{K}_\phi \right\} \geq 1-\theta, \tag{3.13}$$

Proof Since $\{E^{(0)}(C^*, \mathbf{\Gamma}^*), \dots, E^{(k_1)}(C^*, \mathbf{\Gamma}^*)\}$ is a set consisting of i.i.d. random variables, Lemma 1 implies that

$$\Pr\left\{ E^{(0)}(C^*, \mathbf{\Gamma}^*) \leq q \right\} \geq 1-\theta. \tag{3.14}$$

By the definition of $E^{(0)}(C^*, \mathbf{\Gamma}^*)$ in (3.11), this directly implies that $\Pr\left\{\max_{\nu\in\mathcal{K}_\phi} \left(C_\nu^* \|\boldsymbol{e}_\nu^{(0)}(1:N)\| \right) \leq q\right\} \geq 1-\theta$, or $\Pr\left\{\|\boldsymbol{e}_\nu^{(0)}(1:N)\| \leq q/C_\nu,\ \forall \nu \in \mathcal{K}_\phi\right\} \geq 1-\theta$, since the max operator in (3.11) ensures that the bound holds uniformly for all $\nu \in \mathcal{K}_\phi$, completing the proof.

4. Multi-agent Planning

4.1 STL tightening

We recall that a function $f : \mathbb{W} \to \mathbb{R}$ is a Lipschitz continuous if there exists $L \geq 0$ such that for $w_1, w_2 \in \mathbb{W}$,

$$\frac{|f(w_1) - f(w_2)|}{d_{\mathbb{W}}(w_1, w_2)} \leq L < \infty,$$

where $d_{\mathbb{W}}$ denotes a metric on the set $\mathbb{W}$, and L is referred to as the Lipschitz constant. In the sequel, we use the 2-norm on $\mathbb{W}$ to calculate Lipschitz constants, given by $d_{\mathbb{W}}(w_1, w_2) = \sqrt{(w_1 - w_2)^\intercal (w_1 - w_2)}$. We assume the following.

Assumption 1 *(Lipschitz continuity of predicate functions) We assume that all predicate functions appearing in the multi-agent STL formula ϕ are Lipschitz continuous.*

In the following, we tighten the robustness function of the multi-agent STL formula in (2.9) based on the Lipschitz constants of the STL subformulas thereof. First we need the following two Lemmas.

Lemma 2 *For any STL specification ϕ with predicate functions satisfying Assumption 1, the robustness function $\rho^\phi(\boldsymbol{z}(0{:}N)+\boldsymbol{e}(0{:}N))$ is Lipschitz continuous with respect to $\boldsymbol{e}(0{:}N) = (e(0), \ldots, e(N))$, with Lipschitz constant L_ϕ obtained as the maximum Lipschitz constant of the predicate functions appearing in ϕ.*

Proof See [43, Prop. 1].

Lemma 3 *Consider an STL formula ϕ and a trajectory $\boldsymbol{x}(0:N) = \boldsymbol{z}(0:N) + \boldsymbol{e}(0:N)$, where $\boldsymbol{e}(0:N) = (e(0), \ldots, e(N))$, with $e(0) = 0$ and $\Pr\{\boldsymbol{e}(0:N) \in \mathbb{B}(q/C)\} \geq 1 - \theta$. Let L_ϕ be the Lipschitz constant of the robustness function $\rho^\phi(\boldsymbol{z}(0:N) + \boldsymbol{e}(0:N))$ with respect to $\boldsymbol{e}(0:N)$. If $\rho^\phi(\boldsymbol{z}(0:N)) \geq L_\phi \frac{q}{C}$, $\Pr\left\{\rho^\phi(\boldsymbol{x}(0:N)) \geq 0\right\} \geq 1 - \theta$.*

Proof By assumption we have that $\Pr\left\{\|\boldsymbol{e}(0:N)\| \leq \frac{q}{C}\right\} \geq 1 - \theta$. Also, by the Lipschitz condition, it holds that

$$\Pr\left\{\left|\rho^\phi(\boldsymbol{z}(0:N) + \boldsymbol{e}(0:N)) - \rho^\phi(\boldsymbol{z}(0:N))\right| \leq L_\phi \|\boldsymbol{e}(0:N)\| \leq L_\phi \frac{q}{C}\right\} \geq 1 - \theta.$$

Focusing on the case where $\rho^{\phi}(\boldsymbol{z}(0:N)+\boldsymbol{e}(0:N)) \leq \rho^{\phi}(\boldsymbol{z}(0:N))$, the previous Lipscitz inequality yields

$$\Pr\left\{\rho^{\phi}(\boldsymbol{x}(0:N)) \geq \rho^{\phi}(\boldsymbol{z}(0:N)) - L_{\phi}\frac{q}{C} \geq 0\right\} \geq 1-\theta,$$

since $\rho^{\phi}(\boldsymbol{z}(0:N)) \geq L_{\phi}\frac{q}{C}$, completing the proof.

4.2 Relaxation of the chance-constrained synthesis problem

Next, we present the following result, where we relax the chance-constrained synthesis problem in (2.10).

Theorem 1 *Consider the MAS with dynamics in (2.8) subject to the STL formula in (2.9). Let disturbance feedback gains in $\boldsymbol{\Gamma}^* = \{\boldsymbol{\Gamma}_1^*, \ldots, \boldsymbol{\Gamma}_M^*\}$, weights in $C^* = \left\{\{C_{\nu}^*\}_{\nu\in\mathcal{K}_{\phi}}\right\}$ and the quantile parameter q be obtained as in Proposition 2. Denote as $L_{\phi_{\nu}}$, $\nu \in \mathcal{K}_{\phi}$, the Lipschitz constants of the robustness functions of formulas ϕ_{ν}, $\nu \in \mathcal{K}_{\phi}$, Let $\boldsymbol{z}_{\nu}(0:N)$ (and $\boldsymbol{e}_{\nu}(0:N)$), $\nu \in \mathcal{K}_{\phi}$, be nominal (and error) trajectories corresponding to formulas ϕ_{ν}, $\nu \in \mathcal{K}_{\phi}$. Assume*

$$\begin{aligned}
\underset{\boldsymbol{v}(0),\,\boldsymbol{z}(0)}{\text{Minimize}} \; & \sum_{i=1}^{M}\left(\sum_{t=0}^{N-1}(\ell_i(z_i(t),v_i(t))) + V_{f,i}(z_i(N))\right) \\
\text{subject to } & z(t+1) = Az(t) + Bv(t),\; t \in \mathbb{N}_{[0,N)}, \\
& \rho^{\phi_{\nu}}(\boldsymbol{z}_{\nu}(0:N)) \geq L_{\phi_{\nu}}\frac{q}{C_{\nu}},\; \nu \in \mathcal{K}_{\phi},
\end{aligned} \tag{4.1}$$

with $z(0) = x(0)$, has a feasible solution $\boldsymbol{v}(0:N-1) = (v(0),\ldots,v(N-1))$. Then, the multi-agent input sequence $\boldsymbol{u}(0:N-1) = (u(0),\ldots,u(N-1))$, where $u(t) = (u_1(t),\ldots,u_M(t))$, with $u_i(t) = \sum_{k=0}^{t-1}\Gamma_i^{*,t,k}w_i(k) + v_i(t)$, $i \in \mathcal{V}$, $t \in \mathbb{N}_{[0,N-1]}$ together with the corresponding random state trajectory $\boldsymbol{x}(0:N)$ is a feasible solution for (2.10), that is

$$\Pr\{\boldsymbol{x}(0:N) \models \phi\} \geq 1-\theta. \tag{4.2}$$

Proof By the condition in (3.13) we have that $\Pr\left\{\|\boldsymbol{e}_{\nu}(0:N)\| \leq \frac{q}{C_{\nu}},\; \nu \in \mathcal{K}_{\phi}\right\} \geq 1-\theta$, which, by Lemma 3 and the feasibility of the problem in (4.1), leads to $\Pr\left\{\rho^{\phi_{\nu}}(\boldsymbol{x}_{\nu}(0{:}N)) \geq 0,\; \nu \in \mathcal{K}_{\phi}\right\} \geq 1-\theta$, or $\Pr\left(\boldsymbol{x}_{\nu}(0{:}N) \models \phi_{\nu},\; \nu \in \mathcal{K}_{\phi}\right) \geq 1-\theta$, completing the proof.

We remark that to handle STL constraints as in (4.1), we leverage existing STL methods and toolboxes that utilize either integer programming using binary variables or nonlinear solvers using log-sum-exp underapproximations of the STL robustness function. Due to space limitations, we defer a detailed discussion of these technical aspects to an extended version of this work. Next, we decompose (4.1) into individual agent-level problems to improve tractability.

4.3 Distributed control synthesis

We propose an iterative procedure that handles the complexity of (4.1).

4.3.1 Decomposition of STL formula ϕ

For a node i participating in at least one clique, i.e., $i \in \nu$, with $\nu \in \mathcal{K}_\phi$, we define $\mathcal{T}_i$ as the set of cliques containing i by

$$\mathcal{T}_i = \{\nu \in \mathcal{K}_\phi \mid \nu \ni i\}. \tag{4.3}$$

Using (4.3), an equivalent multi-agent STL formula to the original one in (2.9) is defined as $\hat{\phi} = \bigwedge_{i\in\mathcal{V}} \hat{\phi}_i$, where

$$\hat{\phi}_i = \bigwedge_{\nu_i \in \mathcal{T}_i} \phi_{\nu_i}. \tag{4.4}$$

In the following, we denote and define tightened robustness functions as

$$\varrho^{\phi_{\nu_i}}(\boldsymbol{z}_{\nu_i}(0{:}N)) = \rho^{\phi_{\nu_i}}(\boldsymbol{z}_{\nu_i}(0{:}N)) - L_{\phi_{\nu_i}} \frac{q}{C_{\nu_i}},\ \nu_i \in \mathcal{T}_i. \tag{4.5}$$

4.3.2 Agent-level subproblems

We introduce the following problems for the ith agent

$$P_i^0 := \underset{\boldsymbol{v}_i^0, \boldsymbol{z}_i^0}{\text{Minimize}}\ \mathcal{L}_i(\boldsymbol{z}_i^0, \boldsymbol{v}_i^0) \text{ subject to} \tag{4.6a}$$

$$z_i^0(k+1) = A_i z_i^0(k) + B_i v_i^0(k),\ k \in \mathbb{N}_{[0,N)}, \tag{4.6b}$$

$$\varrho^{\phi_i}(\boldsymbol{z}_i^0) \geq 0,\ \text{with } z_i^0(0) = x_i(0), \tag{4.6c}$$

and

$$P_i^t := \underset{\boldsymbol{v}_i^t, \boldsymbol{z}_i^t}{\text{Minimize}}\ \mathcal{L}_i(\boldsymbol{z}_i^t, \boldsymbol{v}_i^t) - \Omega_i \mu_{\nu_t}^t \text{ subject to} \tag{4.7a}$$

$$z_i^t(k+1) = A_i z_i^t(k) + B_i v_i^t(k),\ k \in \mathbb{N}_{[t,N)}, \tag{4.7b}$$

$$\varrho^{\phi_i}(\boldsymbol{z}_i^t) \geq 0,\ \text{with } z_i^t(t) = x_i(t), \tag{4.7c}$$

$$\varrho^{\phi_{\nu_t}}(\boldsymbol{z}_{\nu_t}^t) \geq \mu_{\nu_t}^t,\ \nu_t = \underset{\nu \in \mathcal{T}_i,\ |\nu|>1}{\text{argmin}}\ \{\varrho^{\phi_\nu}(\boldsymbol{z}_\nu^{t-1})\}, \tag{4.7d}$$

$$\mu_{\nu_t}^t \geq \min\left(0, \varrho^{\phi_{\nu_t}}(\boldsymbol{z}_{\nu_t}^{t-1})\right), \tag{4.7e}$$

$$\varrho^{\phi_\nu}(\boldsymbol{z}_\nu^t) \geq \min\left(0, \varrho^{\phi_\nu}(\boldsymbol{z}_\nu^{t-1})\right), \forall \nu \in \mathcal{T}_i \backslash \{\nu_t, i\} \tag{4.7f}$$

where $\Omega_i \gg 0$, $z_i^t(\tau)$ denotes the prediction of $x_i(\tau)$ carried out at time t, $\varrho^{\phi_\nu}(\boldsymbol{z}_\nu^t)$ is the robustness function of the formula ϕ_ν, $\nu \in \mathcal{T}_i$, evaluated over the trajectory $\boldsymbol{z}_\nu^t$, and

$$\boldsymbol{z}_\nu^t = (x_\nu(0), ..., x_\nu(t-1), z_\nu^t(t), ..., z_\nu^t(N)), \tag{4.8}$$

$$\boldsymbol{v}_i^t = (v_i(0), ..., v_i(t-1), v_i^t(t), ..., v_i^t(N-1)), \tag{4.9}$$

$$\mathcal{L}_i(\boldsymbol{z}_i^t, \boldsymbol{v}_i^t) = \sum_{k=0}^{t-1} \ell_i(x_i(k), v_i(k)) + \sum_{k=t}^{N-1} \ell_i(z_i^t(k), v_i^t(k)) + V_{f,i}(z_i^t(N)).$$

In [12], we proposed an iterative procedure for solving a deterministic multi-agent STL problem as in (4.1) via offline synthesis. The procedure initializes at $t = 0$ by solving P_i^0 for all $i \in \mathbb{N}_{[1,M]}$,

generating initial trajectory guesses. For $t > 1$, only a subset of agents $\mathcal{O}_t \subset \mathcal{V}$ update their inputs by solving P_i^t, while the rest retain their previous input sequences. For simplicity, we considered $\mathcal{O}_t$ as a singleton. When $i \in \mathcal{O}_t$, solving P_i^t ensures the satisfaction of the individual task ϕ_i (see (4.7c)), improvement of the most violating (least robust) joint task ϕ_{ν_t} (see (4.7d)-(4.7e)), and non-violation or improvement of other joint tasks (see (4.7f)). The min operator in (4.7e)-(4.7f) relaxes constraints for already satisfied tasks. As shown in [12, Alg. 1], this procedure is recursively feasible, ensures integrity among aggregate trajectories for collaborative tasks, and guarantees either a minimally violating or feasible multi-agent solution within a finite number of iterations. We now extend this approach to distributed online implementation, based on interactions among agents within the same cliques.

4.3.3 Distributed online implementation

Let

$$\boldsymbol{z}_i^t(x_i(t), \boldsymbol{v}_i^t) = (x_i(0), ..., x_i(t), z_i^t(t+1), ..., z_i^t(N)), \tag{4.10}$$

denote a trajectory where the last $N-t$ nominal states are generated by the last $N-t$ inputs of $\boldsymbol{v}_i^t$ starting from $x_i(t)$. At each time $t > 1$, the ith agent computes the robustness functions $\rho^{\phi_{\nu_i}}(\boldsymbol{z}_{\nu_i}^{t-1})$, $\nu_i \in \mathcal{T}_i$, based on the knowledge about $\boldsymbol{z}_{\nu_i}^{t-1} = (x_{\nu_i}(0), \ldots, x_{\nu_i}(t-1), z_{\nu_i}^{t-1}(t), \ldots, z_{\nu_i}^{t-1}(N))$, $\nu_i \in \mathcal{T}_i$, and estimates the robustness function of $\hat{\phi}_i$ as $r_i^t = \min_{\nu_i \in \mathcal{T}_i} \left(\varrho^{\phi_{\nu_i}}(\boldsymbol{z}_{\nu_i}^{t-1}) \right)$. After measuring its state $x_i(t)$, agent-i communicates r_i^t and $\boldsymbol{z}_i^t(x_i(t), \boldsymbol{v}_i^{t-1})$ to all agents in ν_i, $\nu_i \in \mathcal{T}_i$ and receives the corresponding information from them. Agent-i either solves P_i^t if $r_i^t = \min_{j \in \nu_i,\, \nu_i \in \mathcal{T}_i} \left(r_j^t \right)$ or retains its input sequence from $t-1$, that is $\boldsymbol{v}_i^t = \boldsymbol{v}_i^{t-1}$. The overall distributed STL control strategy is summarized in Algorithm 2.

Algorithm 2 Distributed STL control of agent-i

for t in $1 : N$ **do**
  Compute $r_i^t = \min_{\nu_i \in \mathcal{T}_i} \left(\varrho^{\phi_{\nu_i}}(\boldsymbol{z}_{\nu_i}^{t-1}) \right)$
  Measure $x_i(t)$ and $w_i(t-1)$
  Construct $\boldsymbol{z}_i^t(x_i(t), \boldsymbol{v}_i^{t-1})$
  Communicate r_i^t, $\boldsymbol{z}_i^t(x_i(t), \boldsymbol{v}_i^{t-1})$ to $j \in \nu_i$, $\nu_i \in \mathcal{T}_i$
  Receive r_j^t, $\boldsymbol{z}_j^t(x_i(t), \boldsymbol{v}_i^{t-1})$ from $j \in \nu_i$, $\nu_i \in \mathcal{T}_i$
  if $r_i^t < r_j^t$ for all $j \in \nu_i$, $\nu_i \in \mathcal{T}_i$ **then**
    Solve P_i^t and store $(\boldsymbol{v}_i^t, \boldsymbol{z}_i^t)$
  else
    Update $\boldsymbol{v}_i^t \leftarrow \boldsymbol{v}_i^{t-1}$ and $\boldsymbol{z}_i^t \leftarrow \boldsymbol{z}_i^{t-1}$
  Apply $u_i(t) = \sum_{k=0}^{t-1} \Gamma_i^{t,k} w_i(k) + v_i^t(t)$

Proposition 3 *Suppose each agent-i solves P_i^0 at time $t = 0$ and executes Algorithm 2 for $t \geq 1$. Then, for any $1 < t \leq N$, no two agents within the same clique solve P_i^t simultaneously.*

Proof The result follows directly from the selection policy $r_i^t < r_j^t$ for all $j \in \nu_i$, $\nu_i \in \mathcal{T}_i$, which ensures that at most one agent within each clique $\nu_i \in \mathcal{T}_i$ is selected at any given time.

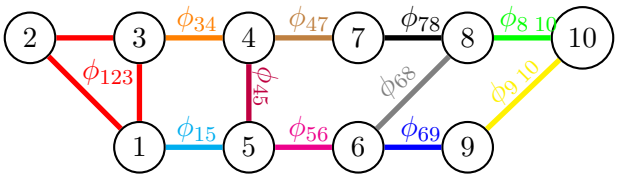

Figure 4.1: The ten agents of the MAS and the cliques $\mathcal{K}_\phi$ in ϕ.

4.4 Example

We note that the following demonstrations were initially presented in the report for the deliverable of WP2 Task 2.4. These results have been adapted to incorporate the new uncertainty quantification method and are reported here for comparison and completeness. We consider ten agents with aggregate dynamics given by $x(t+1) = x(t) + u(t) + w(t)$, where $x(t) = (x_1(t), ..., x_{10}(t)) \in \mathbb{R}^{20}$, $u(t) = (u_1(t), ..., u_{10}(t)) \in \mathbb{R}^{20}$, and $w(t) = (w_1(t), ..., w_{10}(t)) \in \mathbb{R}^{20}$. States $x_i(t) \in \mathcal{X}$, where $\mathcal{X}$ is the workspace confined by the dashed border in Fig. 4.2. Individual inputs are constrained by $\|u_i(t)\|_\infty \leq 0.8$, and disturbances, $w_i(t)$, are Gaussian random vectors, independent time- and agent-wise, with zero mean and covariance, $Q_i = 0.05 I_2$, for all $t \in \mathbb{N}$ and $i \in \mathbb{N}_{[1,10]}$. The MAS is assigned a specification $\phi = \bigwedge_{i=1}^{10} \phi_i \wedge \bigwedge_{\nu \in \mathcal{K}_\phi} \phi_\nu$, with horizon $N = 100$, where $\mathcal{K}_\phi$ is the set of cliques shown in Fig. 4.1, and ϕ_i, ϕ_ν, are tasks assigned to agent i, and the agents in $\nu \in \mathcal{K}_\phi$, respectively. In (2.10), we select $\ell_i(x_i(t), u_i(t)) = \|u_i(t)\|_1$, $V_{f,i}(x_i(100)) = 0$, for all $i \in \mathbb{N}_{[1,10]}$, and set $\theta = 0.70$.

Let $\phi_i = \left(\Box_{[0,100]}(\varphi_i^{\mathcal{X}} \wedge \neg\varphi_i^{O_1} \wedge \neg\varphi_i^{O_2} \wedge \neg\varphi_i^{O_3})\right) \wedge \left(\Diamond_{[10,50]}\varphi_i^{T_i}\right) \wedge \left(\Diamond_{[70,100]}\varphi_i^{G_i}\right)$ be an individual task, which requires agent-i, starting from $x_i(0)$ to pass through T_i and G_i within the intervals $\mathbb{N}_{[10,50]}$ and $\mathbb{N}_{[70,100]}$, respectively, while always staying within $\mathcal{X}$ and avoiding O_1, O_2, O_3. Regions T_i, G_i, $i \in \mathbb{N}_{[1,10]}$, and obstacles O_1, O_2, O_3, are in Fig. 4.2. Note that $x_i(t) \models \varphi_i^{\mathcal{Y}}$ if $x_i(t) \in \mathcal{Y}$, $\mathcal{Y} = \{\mathcal{X}, O_1, O_2, O_3, T_1, \ldots, T_{10}, G_1, \ldots, G_{10}\}$, for $t \in \mathbb{N}_{[0,100]}$.

Let $\phi_\nu = \Diamond_{[0,100]}\left(\|C_\nu x_\nu(t)\|_\infty \leq 1\right)$, where $C_\nu = \begin{bmatrix} I & -I \end{bmatrix}$ if $|\nu| = 2$ or $C_\nu = \begin{bmatrix} I & -I & 0 \\ 0 & I & -I \\ I & 0 & -I \end{bmatrix}$ if $|\nu| = 3$, be a joint task requiring agents in $\nu \in \mathcal{K}_\phi$ (see Fig. 4.1) to approach one another at least once within the horizon.

First, we decompose ϕ according to (4.4), based on the sets $\mathcal{T}_1 = \{(2,3), 5\}$, $\mathcal{T}_2 = \{(1,3)\}$, $\mathcal{T}_3 = \{(1,2), 4\}$, $\mathcal{T}_4 = \{3, 5\}$, $\mathcal{T}_5 = \{1, 4, 6\}$, $\mathcal{T}_6 = \{5, 8, 9\}$, $\mathcal{T}_7 = \{4, 8\}$, $\mathcal{T}_8 = \{6, 7, 10\}$, $\mathcal{T}_9 = \{6, 10\}$, and $\mathcal{T}_{10} = \{8, 9\}$. By selecting sets $\mathcal{O}_k$, $k \geq 1$, as $\mathcal{O}_1 = \{1, 4, 6, 10\}$, $\mathcal{O}_2 = \{3, 5, 7, 9\}$, $\mathcal{O}_3 = \{8, 9, 2, 5\}$, $\mathcal{O}_4 = \mathcal{O}_1$, $\mathcal{O}_5 = \mathcal{O}_2$, $\mathcal{O}_6 = \mathcal{O}_3$, $\mathcal{O}_7 = \mathcal{O}_1$, and so on, we run Alg. 2, which terminates in less than six minutes returning a multi-agent trajectory, illustrated in Fig. 4.2, that satisfies the global STL task ϕ. Fig. 4.3 shows the computational overhead of Alg. 2 vs. a centralized solution for varying agent numbers, using a log scale to highlight the different runtime magnitudes. Note that the runtime of Alg. 2 can further be improved if agent-level subproblems, (4.6), (4.7), can be solved in parallel. By evaluating the robustness function of ϕ for numerous noisy realizations, we see that ϕ is violated in less than 30% of the time, verifying Thm. 1.

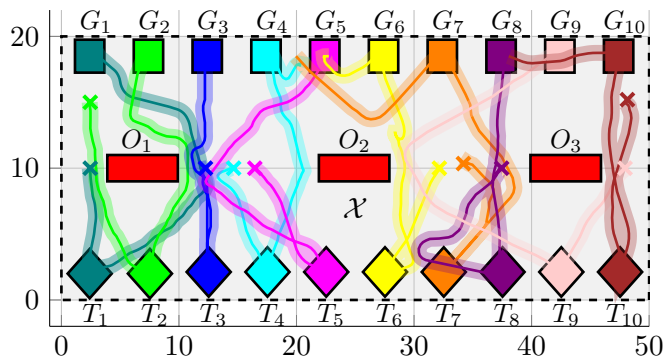

Figure 4.2: Nominal trajectories (solid lines) by Alg. 2, with initial states marked by crosses. Tubes (transparent covers) around trajectories as computed via conformal prediction.

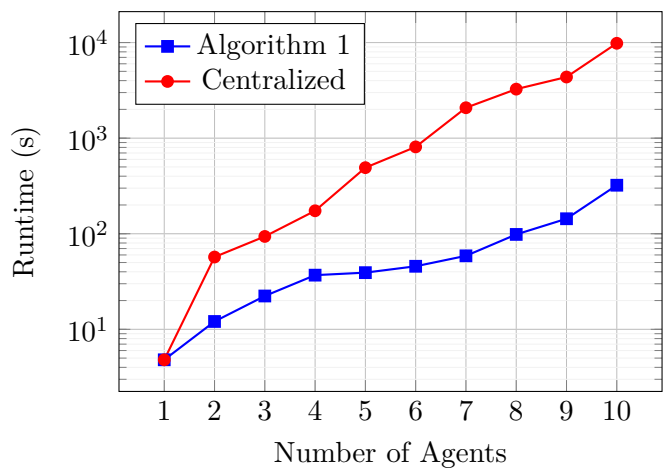

Figure 4.3: Compute times (log scale) for solving (4.1) via Alg. 2 and a centralized approach for varying agent numbers.

5. Conclusion

This report presents a novel approach for multi-agent task planning under uncertainty based on risk awareness, leveraging conformal prediction (CP) to address the challenges of stochasticity and scalability arising in multi-agent systems. By employing CP, we establish a distribution-free framework for data-driven uncertainty quantification and deriving probabilistic guarantees for multi-agent STL task satisfaction. We demonstrate how to transform the chance-constrained optimal control problem into a deterministic optimization problem by tightening the STL robustness function using prediction regions derived from CP and its Lipschitz constants with respect to error trajectories. Furthermore, we introduce an iterative, distributed algorithm that decomposes the large-scale problem into agent-level subproblems, significantly enhancing scalability. This algorithm, guided by a graph-informed scheduling policy, ensures recursive feasibility, guarantees the satisfaction of local tasks, and ensures non-decreasing robustness for joint tasks. The theoretical results highlight the effectiveness of our methodology in handling complex spatiotemporal specifications in stochastic multi-agent systems. This work provides a practical and scalable approach for multi-agent applications under uncertainty and complex collaborative STL specifications, presenting a less conservative alternative to existing methods.

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